25-03-2026

Agenda:

python project — Day 1

- python
- database
- streamlit
- OOP
- data visualization

Issue: doc → pdf

- book
- document
- class notes

(1) scatter
(2) numpy → notion → description →
(3) self healing

AI, npl based

goal → solution of above issue
→ flow of coding
→ OOP

future goals — AI based summary of analytics.

→ how much time user spent in the pdf
↳ % page

→ pdf → collect
→ open → pdf
→ % of pdf →

xyz → 25%

clickstream data ⟶

Clickstream data is the real-time record of a user's activity on a website or app, capturing every click, page view, mouse movement, and scroll behavior in chronological order.

startup ⟶ sell something

analytics ⟶ clickstream data

⟶ add to cart ⟶ Buy

⟶ scrolling ⟶ click product ⟶ add to cart

half hour

high price, low rating

⟶ to understand user behavior

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project

Base design: (Iteration-1)

SD-methodology ⟶ model ⟶ waterfall, agile

upload files ⟶ save file in database
⟶ tags
⟶ description
⟶ date (optional)

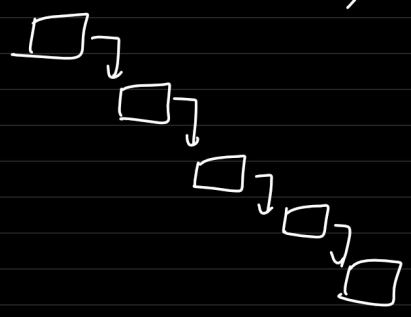
search files ⟶ ~~filename~~
tags OR Date

found ⟶ Yes ⟶ open

Chrome / browser render the pdf

view

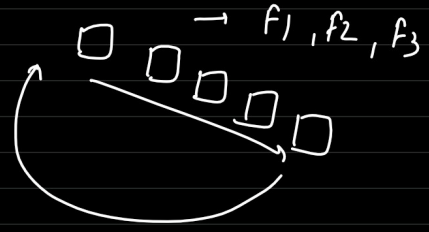
waterfall (f1, f2, f3, f4)



linear sequential

agile

→ f1
→ f2
→ f3
→ f4



iteration process

Base - design (Iteration - 2)

Req gathering → Analytics → User behavior
↙ upload ↓ search ↘ open

Implementation.

Base - design (Iteration - 3)

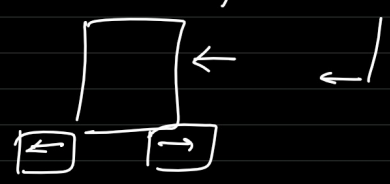
More Analytics → % of pdf → completion rate
→ previous →
→ next →

? how will you track

↙

pages completed / total pages

open - pdf → Adobe Acrobat
pdf view library → research → XYZ library
→ pdf → image → storage (simple implement)



final implementation → analytics

- upload
- search
- open
  - prev
  - next

problem: pdf to hq image → space

↓

40mb → 19b

optimize → [ ] image

pdf → image

Implementation: —

(pdf) upload ← File explorer

↓

object (file)

- save
- file
- name

tags:

description:

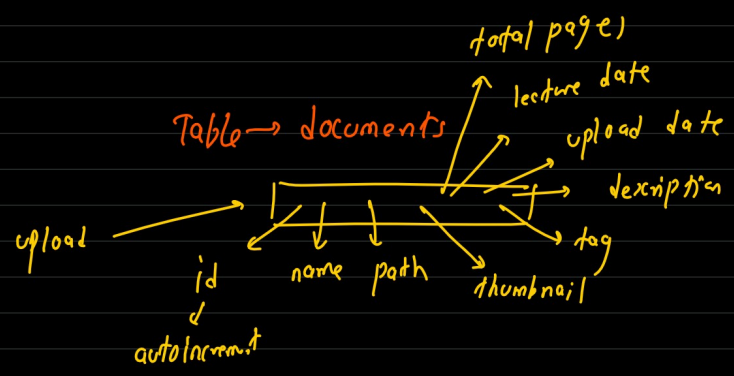
date (optional)

Database

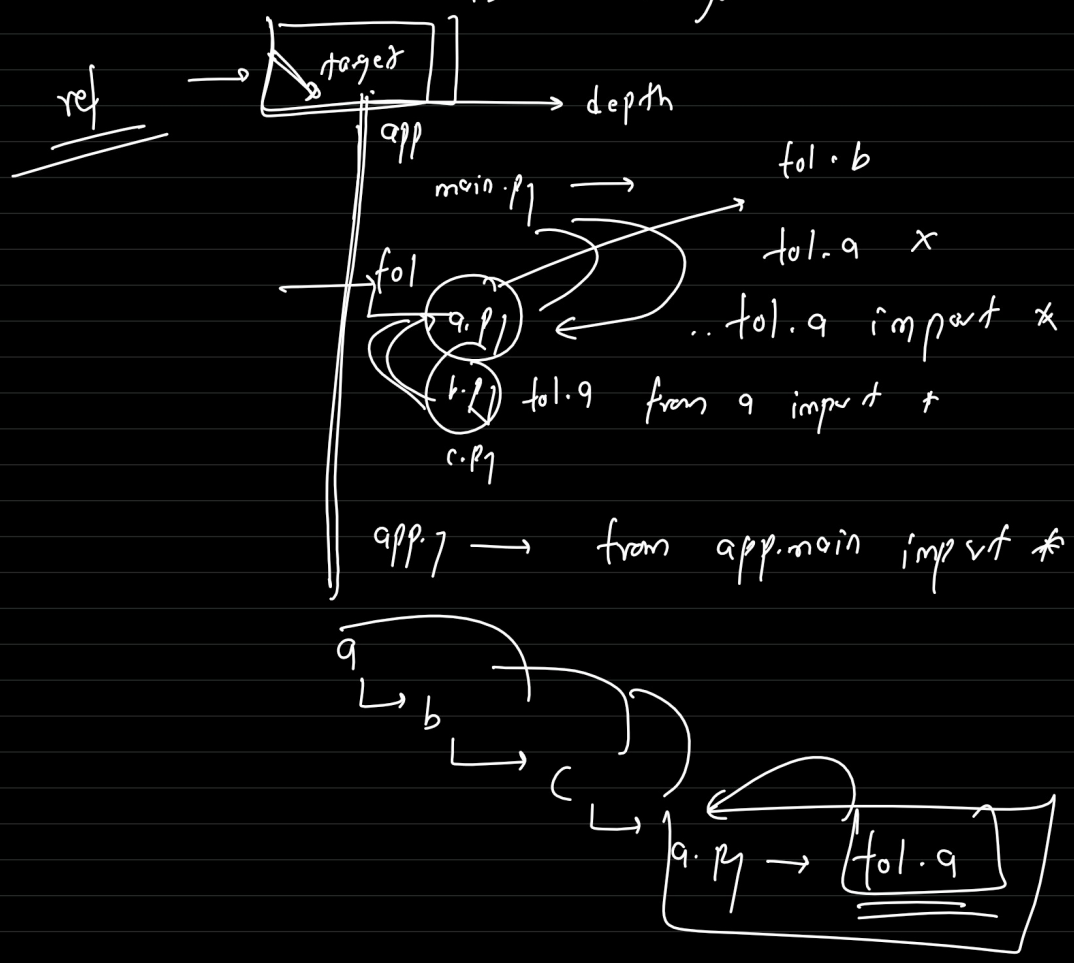
- upload
- id ↓ autoincrement
- name
- path
- thumbnail
- tag
- description
- upload date
- lecture date
- total pages

pdf → imgs → [ ] → low resolution → save ↓ storage

pdf → images → total pages



design decision → db → sqlite

→ db → supabase

→ db → mongodb